Beginner robot factory lesson plan

Build an offline robot factory from ordered instructions. Start with one fixed pickup fixture and one fixed destination. Pair roles: driver edits; observer predicts and checks; swap halfway. Only the instructor enables supervised hardware.

Simulation, paper, reports and measured physical outcomes are separate evidence. Never claim a grasp or placement from a command log alone. Unknown or stale observations require pause and inspection.

Minutes	Duration	Session
0-20	20	Meet our factory
20-45	25	Ordered instructions
45-65	20	Observations and uncertainty
65-95	30	First Python edits
95-105	10	Break
105-120	15	Teach inspect and replay demonstration
120-165	45	Explore three stations
165-210	45	Build the physical sequence
210-230	20	Improve one failure
230-240	10	Three pair demonstrations

Pair rotations

Each 15-minute slot includes 2 minutes to save and transition. Driver edits; observer predicts and checks. Swap roles halfway. One instructor controls the physical integration station.

Explore three stations

Slot	Pair A	Pair B	Pair C
1	Arm and twin observation	Conveyor and observation sequencing	Offline Blockly and Python preparation
2	Conveyor and observation sequencing	Offline Blockly and Python preparation	Arm and twin observation
3	Offline Blockly and Python preparation	Arm and twin observation	Conveyor and observation sequencing

Build the physical sequence

Slot	Pair A	Pair B	Pair C
1	Supervised fixed-fixture integration	Offline sequence rehearsal	Failure tests and evidence
2	Offline sequence rehearsal	Failure tests and evidence	Supervised fixed-fixture integration
3	Failure tests and evidence	Supervised fixed-fixture integration	Offline sequence rehearsal